TrES-2b

R-0092

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_180525` · created 2026-07-15T22:44:55+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458263.9136 +/- 0.00049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1491 +/- 0.0042
Transit depth (Rp/Rs)^2	2.22 +/- 0.12 [%]
Orbital Inclination (inc)	85.34 +/- 0.48
Airmass coefficient 1 (a1)	1.3027 +/- 0.0095
Airmass coefficient 2 (a2)	-0.241 +/- 0.045
Scatter in the residuals of the lightcurve fit is	0.73 %
Best Comparison Star	None
Optimal Aperture	3.96
Optimal Annulus	7.6
Transit Duration (day)	0.0951 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1491 ± 0.0042 vs 0.1254 (deviation 5.05σ)
Duration fit vs published	0.0951 d vs 0.0746 d (deviation 3.99σ)
Mid-time O–C	-1.5 min
Depth SNR	31.8
Residual scatter	0.73 %

Light curve

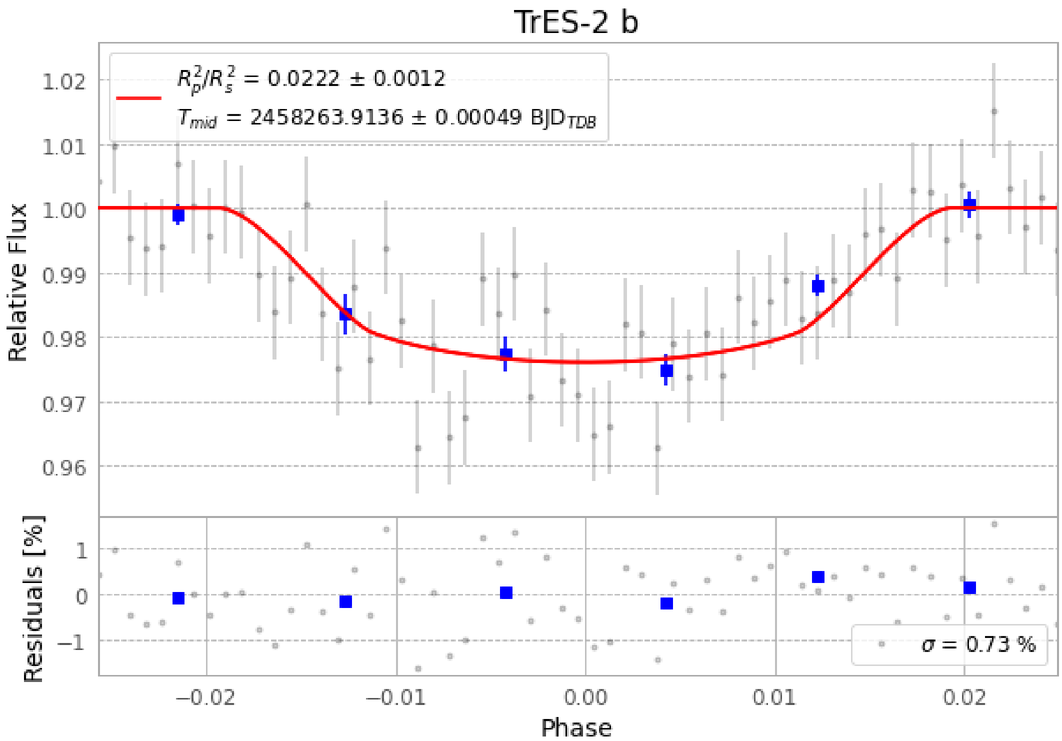

Figure 1 — FinalLightCurve_TrES-2 b_2018-05-25.png (SHA-256 a4aeb3634840bb5b…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [435, 260], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 f398b14fa262d88c…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

61 FITS frames (40 MB), `TRES-2180525082112.FITS` → `TRES-2180525112112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-180525.log (818e28b73544…), FinalLightCurve_TrES-2 b_2018-05-25.png (a4aeb3634840…), FinalLightCurve_TrES-2 b_2018-05-25.csv (933051994698…)

Compute resources

Wall time 141.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)